ISM EXAM CHEAT SHEET

Introduction to Statistical Methods

1. DESCRIPTIVE STATISTICS

Measures of Central Tendency

- **Mean ($\bar{x}$):** $\frac{1}{n}\sum x_i$.
- **Median:** Middle value (sorted). If n is even, average of two middle values.
- **Mode:** Most frequent value.

Measures of Variability

- **Variance (Population σ^2):** $\frac{\sum(x_i-\mu)^2}{N}$
- **Variance (Sample s^2):** $\frac{\sum(x_i-\bar{x})^2}{n-1}$ *(Note the $n-1$!)*
- **Std Deviation (σ, s):** $\sqrt{\text{Variance}}$.
- **Coef. of Variation (CV):** $\frac{s}{\bar{x}} \times 100\%$ (Relative variability).

2. PROBABILITY THEORY

Axioms & Rules

- $0 \leq P(A) \leq 1$ and $P(S) = 1$.
- **Addition Rule:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- **Mutually Exclusive:** $P(A \cap B) = 0$.
- **Independence:** $P(A \cap B) = P(A)P(B)$.

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Read: Probability of A GIVEN B occurred.

Total Probability & Bayes Theorem

If $A_1 \dots A_k$ partition the sample space:

- **Total Prob:** $P(B) = \sum P(B|A_i)P(A_i)$.
- **Bayes Theorem (Posterior):**

$$\boxed{P(A_k|B) = \frac{P(B|A_k)P(A_k)}{\sum_i P(B|A_i)P(A_i)}}$$

Use when: You know $P(Effect|Cause)$ but need $P(Cause|Effect)$.

Naïve Bayes Concept

Assumption: All features X_i are **independent** given class Y.

$$P(Y|X_1 \dots X_n) \propto P(Y) \prod P(X_i|Y)$$

3. RANDOM VARIABLES (RV)

Expectation & Variance

For Discrete RV X with pmf $p(x)$:

- **Expected Value (Mean):** $E[X] = \sum x \cdot p(x)$.
- **Variance:** $Var(X) = E[X^2] - (E[X])^2$.
- **Properties:**
  - $E[aX+b] = aE[X] + b$
  - $Var(aX+b) = a^2 Var(X)$

Two Random Variables (X, Y)

- **Joint Probability:** $p(x,y) = P(X = x, Y = y)$.
- **Marginal Prob:** $p_X(x) = \sum_y p(x,y)$.
- **Covariance:** $Cov(X,Y) = E[XY] - E[X]E[Y]$.
- **Independence Check:** $p(x,y) = p_X(x)p_Y(y)$ for all x, y.

4. DISCRETE DISTRIBUTIONS

Bernoulli Distribution

Single trial (Success/Failure).

- **PMF:** $P(X = x) = p^x(1-p)^{1-x}$ for $x \in \{0, 1\}$.
- **Mean:** p **Var:** $p(1-p)$.

Binomial Distribution

Number of successes in n fixed independent trials.

$$\boxed{P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}}$$

- **Mean:** np
- **Variance:** $np(1-p)$
- **Combo:** $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

Poisson Distribution

Number of events in a fixed interval (time/space).

$$\boxed{P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}}$$

- **Mean:** λ **Variance:** λ
- Use when n is large, p is small ($np = \lambda$).

5. CONTINUOUS DISTRIBUTIONS

Normal (Gaussian) Distribution

$$X \sim N(\mu, \sigma^2)$$

- **Standard Normal (Z):** Convert any X to Z.

$$\boxed{Z = \frac{X - \mu}{\sigma}}$$

- **Why?** To look up probabilities in Z-table.
- **Symmetry:** $P(Z > a) = 1 - P(Z < a)$.

Other Distributions (Intro)

- **t-Distribution:** Like Normal but fatter tails. Use when sample size $n < 30$ and σ is unknown.
- **Chi-Square (χ^2):** Sum of squared standard normal variables. Used in variance testing.
- **F-Distribution:** Ratio of two Chi-squares. Used in ANOVA/Variance comparison.

6. THE "SMART STUDENT" STRATEGY

Identifying the Distribution

Look for keywords in the question:

1. **"...out of 10 items..." / "Success/Failure" → Binomial**. You have n and p.
2. **"...average rate of..." / "...per hour..." → Poisson**. You have λ (rate).
3. **"...normally distributed..." / "Bell curve" → Normal**. Draw the curve, calculate Z-score.

Bayes Theorem Hack

If the question gives you $P(A|B)$ (e.g., "Positive test given Disease") but asks for $P(B|A)$ ("Disease given Positive test"):

- **Step 1:** Draw a Tree Diagram.
- **Step 2:** Calculate Total Probability (Denominator).
- **Step 3:** Apply Formula: $\frac{\text{Branch you want}}{\text{Total Probability}}$.

Common Traps

- **Variance vs Std Dev:** Formulas often give σ, but use σ^2 for calculations. Watch the square root!
- **"At least one":** $P(X \geq 1) = 1 - P(X = 0)$. Much easier to calculate!
- **Sample vs Population:** If calculating variance from data, divide by $n - 1$, not N.